SCALING AUDIO-VISUAL QUALITY ASSESSMENT DATASET VIA CROWDSOURCING

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ABSTRACT

Audio-visual quality assessment (AVQA) research has been stalled by limitations of existing datasets: they are typically small in scale, with insufficient diversity in content and quality, and annotated only with overall scores. These shortcomings provide limited support for model development and multimodal perception research. We propose a practical approach for AVQA dataset construction. First, we design a crowdsourced subjective experiment framework for AVQA, breaks the constraints of in-lab settings and achieves reliable annotation across varied environments. Second, a systematic data preparation strategy is further employed to ensure broad coverage of both quality levels and semantic scenarios. Third, we extend the dataset with additional annotations, enabling research on multimodal perception mechanisms and their relation to content. Finally, we validate this approach through YT-NTU-AVQ, the largest and most diverse AVQA dataset to date, consisting of 1,620 user-generated audio and video (A/V) sequences. The dataset and platform code are available at https://github.com/renyu12/YT-NTU-AVQ.

Index Terms— Audio-visual quality assessment, user-generated content, crowdsourcing, multimodal learning

1. INTRODUCTION

Audio-Visual Quality Assessment (AVQA) provides essential guidance for optimizing multimedia quality by modeling the joint perceptual impact of auditory and visual signals [1]. Unlike Video Quality Assessment (VQA) or Audio Quality Assessment (AQA), AVQA better reflects the comprehensive Quality of Experience (QoE) of subjects when they watch and listen to the A/V sequences.

As is known, data-driven paradigms have already shown remarkable success in signal quality assessment tasks [2, 3]. However, AVQA research progress remains limited due to the lack of datasets. Existing AVQA datasets (representative comparisons in Table 1) remain scarce and generally small-scale [4–7]. They are insufficient to support the training, fine-tuning, and evaluation of advanced data-driven models, let alone systematic studies of human multimodal perception.

The primary challenge lies in scaling AVQA subjective experiments, which typically require controlled in-lab environments and subjects with good auditory discrimination ability, severely limiting dataset size. In contrast, crowdsourcing provides a natural way to scale, but its less controlled environments are particularly problematic for AVQA, where both audio and video must be judged jointly [8]. Nevertheless, prior subjective studies [9–15] have shown that appropriate crowdsourcing experimental design can improve annotation quality, raising the key question: *Can AVQA subjective experiments be effectively scaled through crowdsourcing while maintaining annotation reliability in non-laboratory settings?*

In addition, existing AVQA datasets also suffer from a lack of diversity in terms of audio-visual content, distortion types, and quality distributions. Thus, they fail to capture the complexity of real-world user-generated content, which restricts their practical value and limits the generalizability of trained models. Furthermore, their annotations are typically limited to overall quality scores. It constrains the exploration of how different modalities and semantic factors shape human multimodal quality perception.

To address these 3 limitations, We propose a scalable approach for AVQA dataset building as follows:

- We design the first crowdsourced subjective experiment framework for AVQA, enabling large-scale data annotation with reliable quality in non-laboratory environments through checks on experiment procedure, rating results and subjects.
- We propose a stratified sampling and manual selection method to widely select distorted A/V sequences from open-access resources, enabling coverage of diverse distortion types and semantic scenarios. This results in YT-NTU-AVQ, the largest AVQA dataset to date of 1,620 user-generated A/V sequences.
- We extend the dataset with rich annotations, including modality-specific quality, attention weights, audio and video categories, and video summaries. This provides resources to study multimodal perception mechanisms and their dependence on semantic scenarios.

Dataset	Source	Collection	Unique AV	Total AV	Distortion	Resolution	Length	Label	Rating Method	Year
UnB-AVQ	CDVL	Manual	6	72	Synthetic	720p	8s	Multi-MOS	In-Lab	2013
LIVE-SJTU	CDVL	Manual	14	336	Synthetic	1080p	8s	AVQA MOS	In-Lab	2019
SJTU-UAV	YFCC100M	Manual	520	520	In-the-wild	720p–1080p	8s	AVQA MOS	In-Lab	2022
Ours	YouTube CC	Sampling + Manual	1620	1620	In-the-wild	≤1080p	10s[†]	Multi-label	Crowdsourcing	2025

Table 1: Comparison of representative AVQA datasets. ([†]A small fraction of YT-NTU-AVQ sequences are shorter than 10s)

2. SUBJECTIVE EXPERIMENT FRAMEWORK

Crowdsourcing-based AVQA experiments provide an efficient alternative to traditional lab-based studies by removing physical constraints and enabling large-scale participation from diverse environments. However, AVQA-specific crowdsourcing remain largely unexplored due to unique challenges:

- Varied experimental conditions: Subjects use various devices with different display and audio characteristics, which cannot be fully regulated.
- Inconsistent scoring standards: Subjects may interpret the rating task differently without intensive training.
- Unreliable submissions: A proportion of annotations may be careless, random, or even fraudulent.

Despite these issues, we argue that a carefully designed crowdsourcing experiment is both feasible and valuable for AVQA, because a large and diverse subject pool reduces random variability and improves the representativeness of collected opinions. Our goal is to capture general user perceptions under realistic conditions.

In addition, although these challenges cannot be fully resolved, they can be mitigated: basic environment control address varied experimental conditions, simple training reduces inconsistent interpretations of the task, appropriate filtering removes unreliable submissions, and subjects screening ensures the inclusion of reliable subjects.

Therefore, we develop a practical framework including a series of measures across experimental procedure, result filtering, and subject screening to improve the reliability and efficiency of subjective annotation.

2.1. Experiment Platform Design

we developed a custom platform based on jsPsych [16]. It can support the complete experiment session: environment check, consent, instruction, training, and quality rating. Key features include: (1) enforcing environment requirements and playback restrictions, (2) logging user interactions to detect inattentive behavior, (3) providing brief instructions and five training videos, and (4) collecting multiple subjective annotations.

For environment control, we adopt methods such as soft confirmation (quiet space, proper screen, headphones), limited automatic checks (device, resolution, audio output, network), and enforced playback restrictions (full-screen mode, no timeline dragging, mandatory completion before submission).

To study multimodal perception, we collect multidimensional ratings rather than a single overall score. In a crowdsourcing setting, however, subjects' attention is limited, so the number and complexity of questions must be carefully controlled. 4 questions in our experiment are as follows (single-stimulus with discretized slider ratings: Q1–Q3: 1.0–5.0, step 0.1; Q4: 0–100%, step 1%):

- **Q1 (AVQA Score) :** *Rate overall audio-visual quality.*
- **Q2 (AV_VQA Score) :** *Rate video quality only.*
- **Q3 (AV_AQA Score) :** *Rate audio quality only.*
- **Q4 (Weight of AV) :** *Which Part do you pay more attention to when you evaluate the overall quality? (Give a 100% split, e.g., Audio 50% : 50% Video).*

2.2. Ranking-based Data Filtering

A pilot study was conducted to validate the feasibility of our crowdsourcing design, but it also revealed a non-negligible amount of low-quality results. We tested different conventional filtering methods, but still struggled to identify a large portion of unreliable submissions, which often appeared as randomly distributed mid-range ratings.

Motivated by these failure cases, we propose a simple yet effective dynamic filtering approach combining **ranking consistency** (SROCC with group consensus) and **score dispersion** (standard deviation). Ranking consistency filters out randomized responses that break the overall ordering of videos, while score dispersion removes overly uniform or concentrated ratings. Only results exceeding both thresholds are considered reliable. This joint criterion reliably identifies uninformative or careless submissions, substantially improving rating quality.

2.3. Multi-stage Experiment

Relying solely on data filtering after the experiment is inefficient and costly, since many unreliable submissions may already be collected. We observed that submission quality strongly depends on subject reliability, motivating a three-stage framework to screen subjects.

In brief, we first build a small reference subset through the pretest, then dynamically screen new subjects in the qualification test, and finally let only qualified subjects rate the

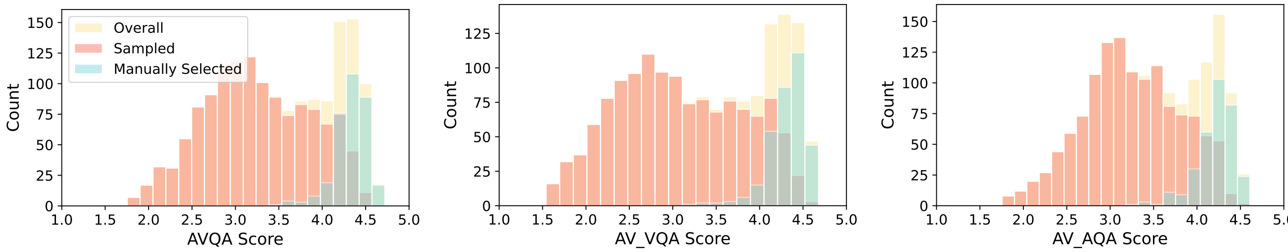


Fig. 1: Distributions of MOS scores in our AVQA dataset. From left to right: overall audio-visual quality (AVQA), video-only quality (AV_VQA), and audio-only quality (AV_AQA). (Mean and standard deviation: AVQA: $\mu = 3.47$, $\sigma = 0.72$; AV_VQA: $\mu = 3.49$, $\sigma = 0.77$; AV_AQA: $\mu = 3.44$, $\sigma = 0.64$)

remaining set of videos in the formal test. This design enables dynamic and scalable subject screening, ensuring that large-scale data collection remains both reliable and efficient.

The details in our experiment are as follows. **Pretest:** 120 videos are used to establish reference scores and identify reliable subjects via ranking-based filtering. **Qualification test:** new subjects rate 30 videos from the pretest subset and are qualified only if they meet the filtering criteria. **Formal test:** only qualified subjects proceed to rate the remaining 1,500 videos. Each group contains 30 videos to keep an acceptable session duration, and all results are ultimately rechecked with ranking-based filtering to discard low-quality data.

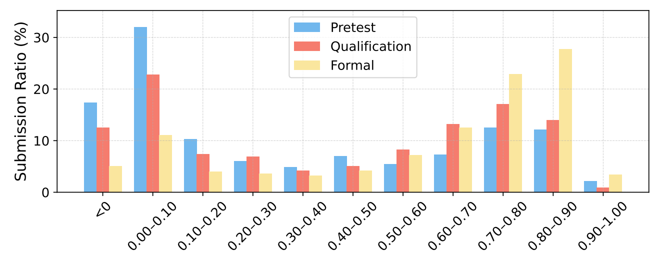


Fig. 2: Distribution of average SROCC values across submissions in different crowdsourcing stages. Submission quality improves progressively from pretest to formal evaluation, validating the effectiveness of our multi-stage filtering pipeline.

3. DATA PREPARATION

Dataset diversity is essential for the generalization of AVQA models. Yet existing AVQA datasets rely on manual selection, which severely limits diversity and scalability. A better strategy is to first construct a large candidate pool and then apply stratified sampling, with limited manual selection used for supplementation. But a major challenge is **audio-relevant sequence selection**. Unlike VQA, where clips can be randomly cropped, AVQA requires segments with meaningful or salient audio, which are unevenly distributed and often sparse.

A feasible strategy is to combine audio event detection models with manual checking, while we adopt a simpler solution by leveraging the existing audio-visual understanding dataset VALOR [17], which already provides over 1M 10-second YouTube A/V sequences with rich audio-visual content and semantic labels.

From this pool, we applied a stratified sampling strategy to maintain coverage across multiple factors. Three attributes were prioritized: (1) **audio quality**, using pseudo-labels from AudioBox [18]; (2) **video quality**, predicted by Faster-VQA [19]; and (3) **audio semantics**, obtained by coarsely mapping AudioSet [20] labels into speech, music, and sound with seven combinations. We first sampled 10k candidates, then selected 1,296 while reweighting attribute distributions to broaden label coverage.

Another practical issue is that many sequences from existing sources [21, 22] are outdated, with most videos published before 2015. To address this, we manually added 324 CC-licensed sequences from post-2017 YouTube videos, ensuring recency and topical diversity. Together, these steps yield a dataset of 1,620 clips with balanced coverage of semantic classes and quality ranges.

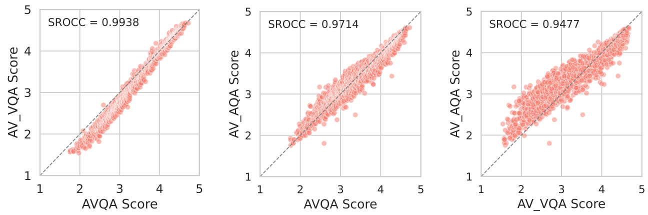
In addition, we complemented missing semantic annotations for all sequences with Gemini2.5 [23] predictions and manual verification, covering audio and video categories, and video summaries.

4. RESULTS AND ANALYSIS

4.1. Subjective Experiment Results

All experiments were conducted on Amazon Mechanical Turk (AMT) with subjects required to have more than 500 completed tasks and an approval rate above 97%. After applying filtering criteria ($SROCC > 0.5$, $STD > 0.5$), the three stages yielded the following:

Pretest. 328 tasks for 120 videos produced 9,840 ratings from 286 subjects, of which 3,750 (31.3 per video, 38.1% acceptance) were retained as reliable references. **Qualification.** 1,952 new participants rated one pretest group each, yielding 32,490 valid ratings (270.8 per video, 49.1% acceptance) and qualifying 1,051 subjects. **Formal.** For the remaining 1,500 videos, 4,680 tasks produced 94,590 reliable ratings from 580 qualified subjects (63.1 per video, 67.4% acceptance).



(a) (b) (c)

Fig. 3: Scatter plots of MOS comparisons: (a) AVQA vs AV_VQA, (b) AVQA vs AV_AQA, and (c) AV_VQA vs AV_AQA. All three pairs show high correlations, revealing strong monotonic relationships among the score types.

Type	Method	SROCC	PLCC
AQA	PAM	0.4217	0.4358
	AudioBox (CE)	0.4328	0.4335
	AudioBox (PQ)	0.5145	0.5335
VQA	FasterVQA	0.7868	0.7865
	FastVQA	0.8503	0.8514
	Q-Align	0.9342	0.9329
AVQA	BRISQUE+MFCC	0.4078	0.4233
	V-BLIINDS+eGeMAPS	0.6525	0.6630
	VRN-50+ARN-50	0.8425	0.8412
	DNN-RNT	0.8383	0.8324
	GeneralAVQA	0.9337	0.9380

Table 2: Performance (SROCC/PLCC) of baseline models on YT-NTU-AVQ dataset.

Figure 1 presents the MOS distributions of the overall AVQA score, video-only quality score (AV_VQA), and audio-only quality score (AV_AQA). The sampled subset exhibits approximately Gaussian-shaped quality distributions, indicating the effectiveness of our stratified sampling. The manually selected subset further supplements the high-quality range, ensuring broader coverage of quality.

Figure 2 shows clear improvements in submission reliability across stages, confirming the effectiveness of our experiment framework.

4.2. Analysis into Multimodal Quality Perception

All three subjective scores (AVQA, AV_VQA, AV_AQA) show extremely high correlations, indicating that unimodal ratings are not strictly independent but largely shaped by overall perception. In particular, the near-perfect alignment between AVQA and AV_VQA suggests that "visual quality dominates" in AVQA for in-the-wild UGC content.

This is consistent with our baseline results (Table 2), VQA models achieve surprisingly high performance. We evaluate YT-NTU-AVQ using representative AQA, VQA, and AVQA models. Pretrained AQA models (PAM [24], AudioBox [18])

Group	n	Subj. A%	Diff (V)	Diff (A)
A << V	87	56.95	-0.06	+0.37
A < V	284	54.20	+0.01	+0.18
A ≈ V	489	51.97	+0.08	+0.09
A > V	291	48.52	+0.20	+0.01
A >> V	469	45.08	+0.34	-0.17
Overall	1620	50.01	-0.02	+0.05

Table 3: Subjective audio attention and AVQA–modality score differences across groups defined by the relative quality difference $AV\_AQA - AV\_VQA$ (thresholds: ± 0.1 slight, ± 0.3 large). **Subj. A%**: mean subjective audio attention. **Diff (V/A)**: mean difference AVQA minus video/audio scores (positive indicates AVQA score is higher).

and VQA models (FasterVQA [19], FastVQA [25], Q-Align [26]) are directly applied to the entire dataset. For AVQA, we evaluate fusion of A/V features by SVR [6] with representative feature combinations [7, 27–30] and deep learning-based models (DNN-RNT [31], GeneralAVQA [7]) using 5-fold cross-validation.

A reasonable explanation is that audio quality in UGC is often concentrated in mid-to-high range with few severe distortions, and humans are less sensitive to subtle audio differences, making video variations appear more influential in quality assessment [32–35]. As a result, this apparent visual quality dominance is expected to be stronger in UGC datasets.

An interesting observation arises from the subjective attention weight. As shown in Table 3, subjects report nearly balanced audio attention on average ($\mu = 50.01$ and $\sigma = 4.32$). However, when the quality scores of the two modalities differ, attention tends to shift toward the degraded modality, while the overall AVQA score remains closer to the higher-quality modality. This asymmetry suggests that humans focus on distortions but anchor their judgments by the better modality. Another important factor is semantic context: in scenarios where audio is important and highly correlated with video (music performance, dance, and people speaking), the reported audio attention is lower than in other scenarios, suggesting a counterintuitive holistic audio-visual integration.

5. CONCLUSION

We proposed the first crowdsourced subjective experiment framework tailored for AVQA, demonstrating that reliable quality assessment can be achieved even under non-laboratory settings. Combined with a designed data preparation strategy, this framework enabled the construction of YT-NTU-AVQ, the largest and most diverse AVQA dataset with multi-dimensional annotations. The dataset validates the feasibility of scaling AVQA beyond lab settings and provides a useful resource for future research as well as for understanding human multimodal quality perception.

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Supplementary Material

This supplementary material focuses on implementation details of the experimental platform, crowdsourcing setup, and dataset construction. It is intended to support reproducibility and facilitate future dataset extension or similar experiment design.

A. EXPERIMENTAL PLATFORM

A.1. Environment Check

Before starting the rating session, subjects are guided through an environment preparation and hardware verification stage. In web-based crowdsourcing experiments, strict control over the testing environment is not feasible. Therefore, we provide clear preparation instructions and perform lightweight automatic checks to reduce the risk of unreliable experimental conditions.

Subjects are first asked to confirm recommended conditions (Figure 4), including a quiet environment, use of a desktop device with a sufficiently large screen, headphone-based listening, and stable network connectivity. If the subject indicates that the environment is not ready, the experiment cannot proceed.

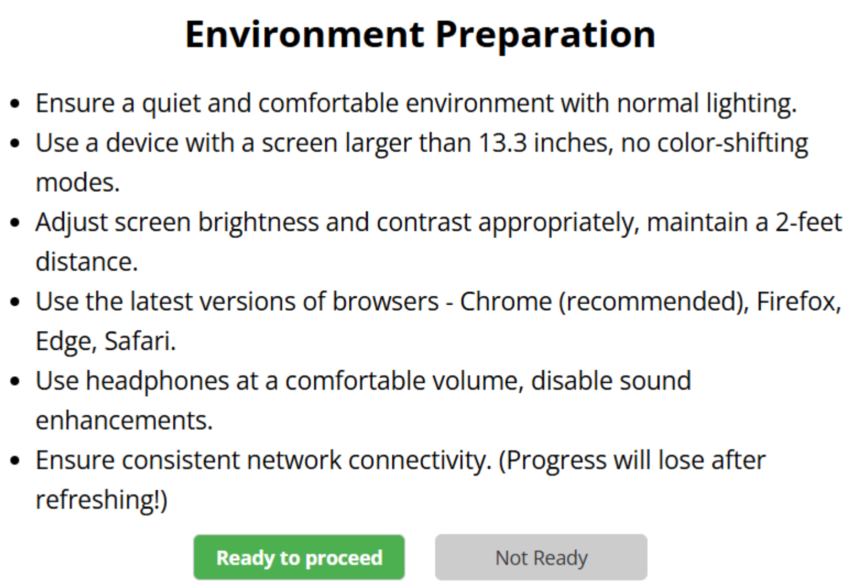


Fig. 4: Screenshot of the Environment Preparation Confirmation Page

The platform then performs basic automatic checks using browser APIs, including:

- **Display capability:** verifying that the effective screen resolution is larger than 720p. (Enforcing a strict 1080p requirement is unreliable in web environments due to variations in browser reporting, device pixel ratio, and operating system scaling)
- **Network connectivity:** performing a short media download test to ensure stable video loading.
- **Audio output availability:** verifying that an audio output device is present.

If any check fails, subjects are allowed a limited number of retries. If all retries fail, the experiment session is terminated.

In addition, the User-Agent string is recorded during task submission to help identify the client device type and to detect submissions from unsupported device categories (e.g., mobile devices).

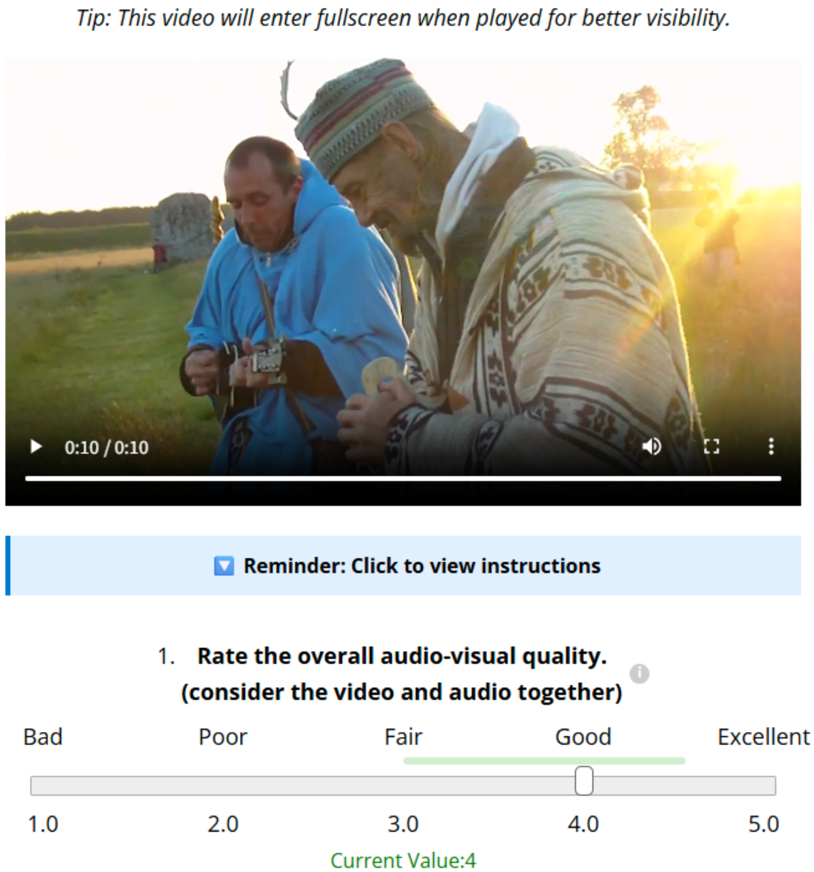


Fig. 5: Screenshot of the A/V Sequence Rating Interface

A.2. A/V Sequence Rating Interface

In each experimental session, subjects consecutively rate 30 A/V sequences. Upon entering the page shown in Figure 5, the video automatically starts in full-screen mode. During the first playback, any attempt to drag the progress bar triggers a warning, and submission is not allowed unless the total watch time meets or exceeds the video duration. After the initial playback, subjects may freely replay the video as many times as they wish. Throughout the experiment, each video is preloaded to minimize network-induced stuttering.

Following each video, subjects provide ratings using a single-stimulus evaluation interface with four slider-based questions. The primary rating question asks for an overall audio-visual quality score on a continuous scale from 1 to 5 with a step size of 0.1. The slider initially appears at the leftmost position and must be moved before submission is enabled.

The interface also provides a collapsible Instruction panel and on-hover descriptions for each question. During the training phase, an expected score range (displayed as a green bar) is shown as a broad guideline; subjects are required to adjust their sliders within this range to proceed, ensuring they become familiar with the approximate rating standard.

To further explore the mechanisms underlying human perception of audio-visual quality, three supplementary questions (Q2–Q4) were added to the interface, all answered via slider controls. Q2 and Q3 use the same 1–5 scale as Q1, while Q4 captures subjects' perceived attention weights between audio and video, represented as a continuous allocation ranging from 0%:100% (audio:video) to 100%:0%, with a step size of 1%.

A.3. Task Assignment and Submission

We deployed a custom backend server to manage task distribution and data collection. The subject's AMT worker ID is transmitted to the server when requesting a task, allowing the system to check submission history and prevent duplicate task

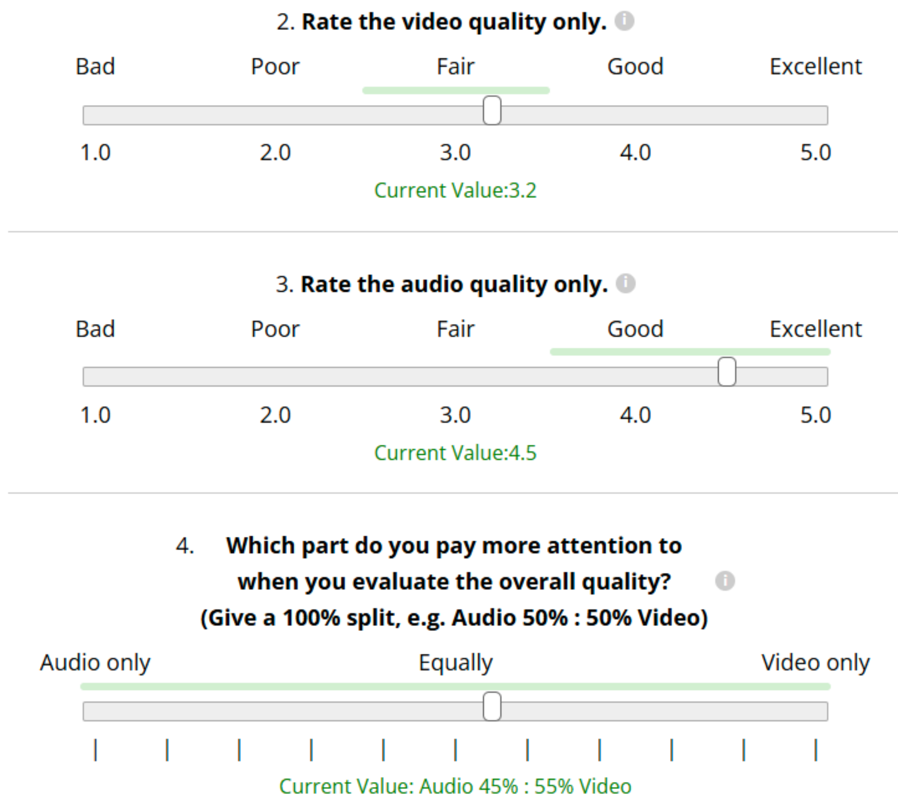


Fig. 6: Screenshot of the Additional Questions

assignments. All experimental videos are pre-grouped in advance. When a subject requests a task, the server randomly assigns one unfinished task group based on the subject's task history.

After completing the experiment, all rating results are transmitted to the backend server and stored for subsequent statistical analysis. Upon successful submission, the server generates a random completion code, which is returned to the subject for confirmation of task completion on the AMT platform.

This design ensures balanced task distribution across video groups and reduces the risk of repeated submissions, preventing certain videos from being oversampled and helping maintain a more uniform number of ratings per video.

B. CROWDSOURCING EXPERIMENT

B.1. Qualification Settings

To ensure high-quality participation, we adopted a multi-stage qualification strategy. The Formal Test stage was restricted to experienced and reliable workers who satisfied all predefined criteria.

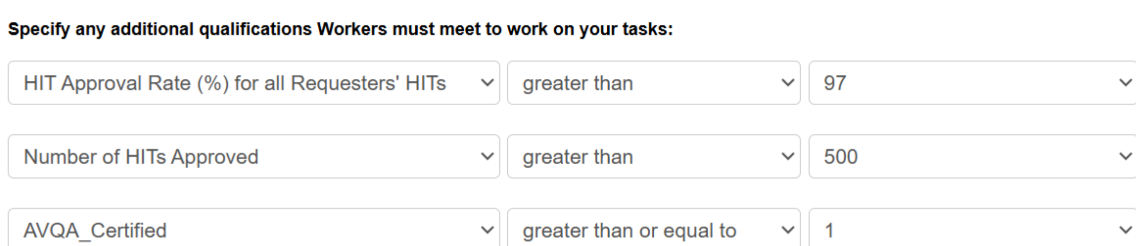


Fig. 7: Qualification requirements for workers to participate in the Formal Test on AMT.

As shown in Figure 7, three qualification filters were enforced for the Formal Test:

HIT Approval Rate: Workers were required to have an approval rate above 97% for all previously completed tasks.

Number of HITs Approved: Workers needed to have successfully completed more than 500 tasks.

AVQA_Certified Qualification: This custom qualification was only granted to workers whose submissions in both the Pretest and Qualification Test stages passed the dynamic filtering procedure described in Section B.2.

Here, a HIT (Human Intelligence Task) refers to an individual task posted on the AMT platform. Each HIT in our study corresponds to a single experimental session where the subject rates a set of videos.

To streamline qualification management, we leveraged the AMT API to automatically update qualifications in bulk. Upon verifying a worker's eligibility, our backend system issued the AVQA_Certified flag programmatically, ensuring that only qualified subjects could access the Formal Test without manual intervention.

B.2. Dynamic Filtering of Subjective Data

To ensure the reliability of crowdsourced ratings, we designed a dynamic filtering method based on rank correlation and score variance. First, all invalid submissions were removed. We then aggregated the ratings of all valid submissions to compute the Mean Opinion Score (MOS) for each video. Using these preliminary MOS values as the reference, we evaluated the quality of each individual submission by calculating:

SROCC (Spearman Rank-Order Correlation Coefficient) between the subject's ratings and the aggregated MOS, computed separately for the three rating dimensions (AVQA, AV_VQA, AV_AQA) and then averaged.

STD (Standard Deviation) of the subject's ratings across the three rating dimensions, also averaged.

Submissions that achieved both an average SROCC and average STD above the thresholds (0.5, 0.5) were retained, and the MOS was recomputed using only the filtered ratings to obtain the final subjective scores.

```
Algorithm: Dynamic Filtering of Crowdsourced Ratings
1. Remove duplicate and invalid submissions.
2. Compute preliminary MOS for each video.
3. For each submission s:
   a. Merge s with video MOS labels.
   b. Calculate SROCC(s, MOS) for AVQA, AV_VQA, AV_AQA.
   c. Calculate STD(s) for AVQA, AV_VQA, AV_AQA.
   d. Compute avg_SROCC and avg_STD.
4. Keep s if avg_SROCC > 0.5 and avg_STD > 0.5.
5. Recompute final MOS using only valid ratings.
```

In principle, preliminary MOS can be recomputed either after each round of filtering or when new crowdsourced data are added. However, in practice we observe that MOS estimates remain relatively stable, and therefore iterative refinement is not necessary.

The above filtering procedure was applied consistently across all stages of the crowdsourcing experiment. The resulting data retention statistics for each stage are summarized below.

Pretest. We released 328 tasks for 120 videos (4 groups). From 286 subjects, 9,840 ratings were collected. After dynamic filtering, 125 valid tasks remained, producing 3,750 reliable ratings (31.25 per video, 38.1% acceptance). These scores served as initial references for qualification.

Qualification Test. 1,952 new subjects rated one pretest group each. Dynamic filtering retained 958 (49.08%), and together with 93 pretest-qualified subjects, 1,051 subjects were qualified. Final MOS for the 120 videos was recomputed using 32,490 accepted ratings (270.75 per video).

Formal Test. For the remaining 1,500 videos, 4,680 tasks were launched. From 580 qualified subjects, 3,153 tasks passed filtering, yielding 94,590 reliable ratings (63.06 per video, 67.37% acceptance).

C. DATASET DETAILS

C.1. Sampling Strategy

To construct a representative subset under limited dataset size, we sample sequences to balance key audio-visual attributes while preserving real-world UGC characteristics. Although many attributes (e.g., perceptual quality, low-level signal statistics,

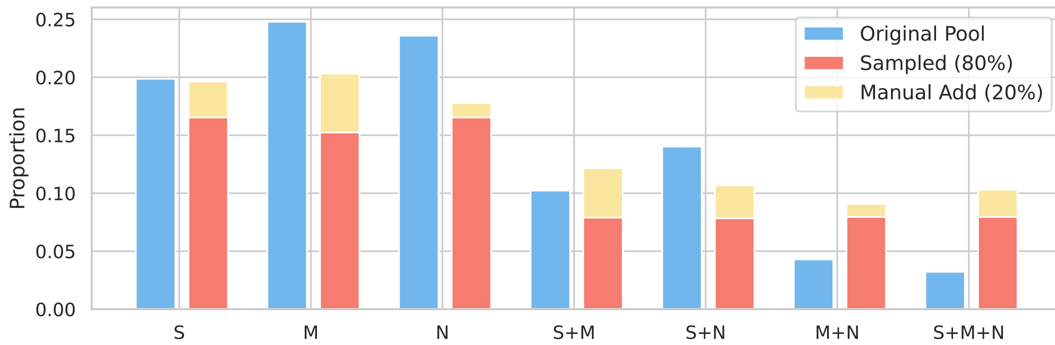


Fig. 8: Distribution of audio semantic label combinations after sampling.

semantic categories) can be considered, many are intrinsically correlated. Enforcing uniform coverage across all attributes is impractical due to data sparsity. Therefore, we prioritize three key attributes with a focus on audio to support deeper analysis.

Audio quality. We use AudioBox to generate pseudo-labels and adopt its Clarity Estimation (CE) and Perceived Quality (PQ) metrics to capture perceptual degradations across diverse audio types.

Video quality. We adopt FasterVQA to provide reliable video quality predictions while maintaining computational efficiency.

Audio semantic category. We map AudioSet labels into three coarse categories (speech, music, sound) and consider their seven label combinations to improve semantic coverage.

We adopted a stratified sampling approach that jointly considers audio category combinations and pseudo-quality scores for audio and video. Discrete audio labels were balanced according to predefined ratios (with single categories weighted as 2 and combined categories weighted as 1), while continuous features (pseudo-quality) were softly equalized by binning and applying a smoothing coefficient. This process slightly increases the sampling probability of underrepresented bins without heavily distorting the original distribution. The balancing strength was controlled by a coefficient $\alpha = 0.3$, and continuous features were discretized into $n = 8$ bins. The parameter α adjusts the degree of equalization: when $\alpha = 0$, no balancing is applied; larger α values lead to stronger flattening of the bin distribution.

```
Algorithm: Stratified Sampling with Soft Balancing
1. Define target ratios for discrete audio label groups.
2. For each group:
   a. Discretize continuous pseudo-quality features into n_bins (n=8).
   b. For each bin i, compute original probability p_i.
   c. Define uniform distribution u_i = 1 / n_bins.
   d. Calculate bin weight w_i = (u_i / p_i)^alpha, normalize to sum=1.
   e. Assign each sample a weight according to its bin's w_i.
   f. Sample target_n videos proportionally to the adjusted weights.
3. Merge all sampled groups to form a balanced set of 10,000 videos.
4. Randomly select 1,296 videos from the 10,000 for the final dataset.
```

Using this approach, we first sampled 10,000 videos from the VALOR-1M pool with improved coverage of labels and quality ranges. From this set, 1,296 videos were randomly selected as the final dataset for subjective experiments.

The resulting distribution of audio semantic combinations after sampling is shown in Figure 8.

The distributions of pseudo labels before and after sampling are shown in Figure 9, including audio quality based on AudioBox predictions, and video quality based on FasterVQA predictions.

C.2. Manual Search Procedure

To complement automated sampling, we performed manual video selection to ensure diversity and content quality. Following the category design of prior datasets such as YouTube SFV+HDR, we defined ten categories: **Animal, Cooking, Dance, Gameplay, Health, Hobby, Music, Travel, Speech**, and **Sports**. For each category, more than thirty keywords were generated with the assistance of ChatGPT and manually refined to guide the search process.

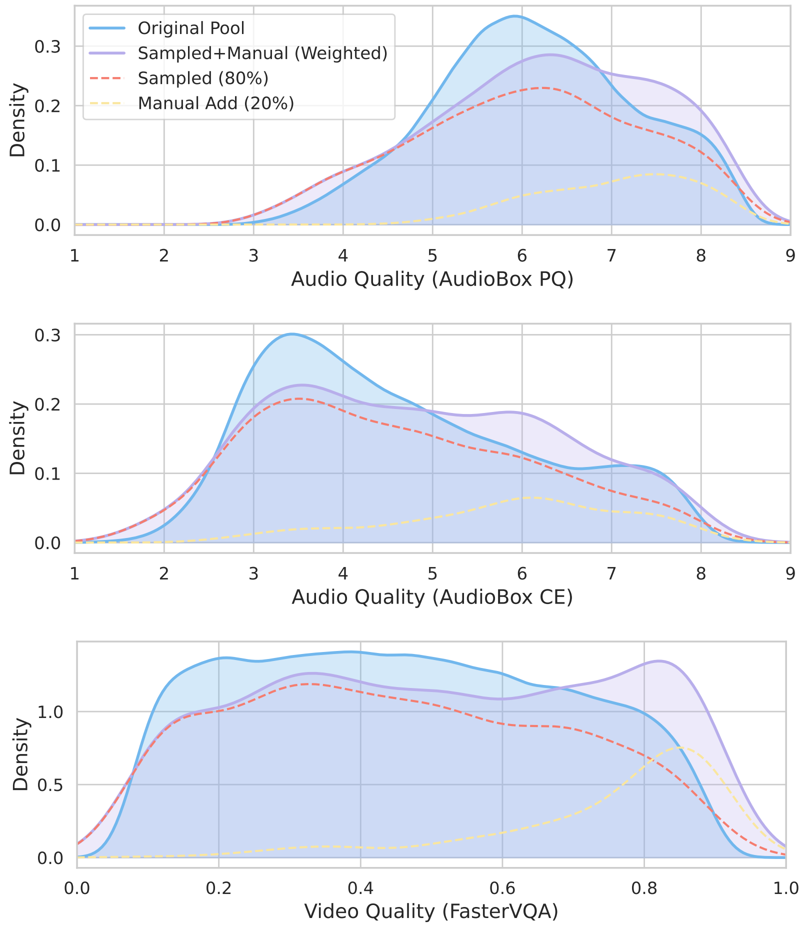


Fig. 9: Distributions of pseudo labels before and after sampling. From top to bottom: audio quality (AudioBox PQ), audio quality (AudioBox CE), and video quality (FasterVQA). Weighted sampling and manual additions improve label coverage across the score range.

Candidate videos were retrieved from YouTube under Creative Commons licenses and manually screened to ensure:

- relevance to the intended category with diverse scenes and creators;
- sufficient visual quality and recent upload time (preferably within the last two years);
- landscape orientation and appropriate content;
- exclusion of unsuitable material, such as re-uploaded or disturbing content.

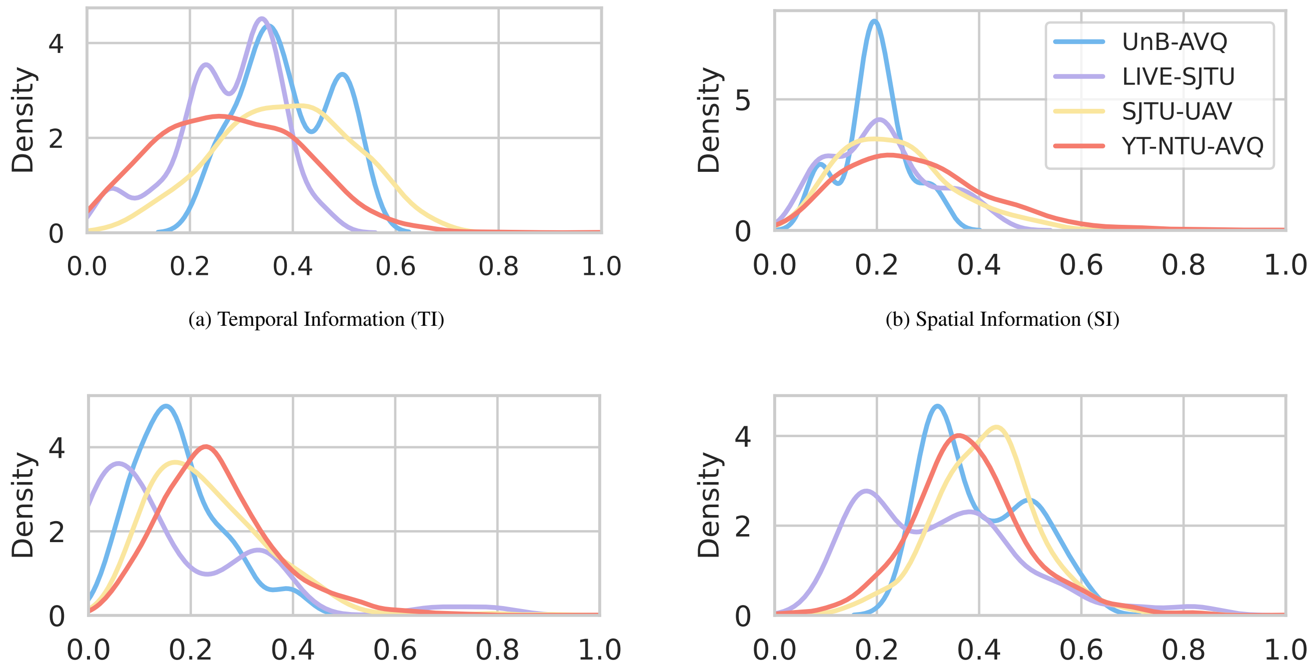
For each keyword, at least the top 50 search results were reviewed, and eligible videos were selected to supplement the final pool.

C.3. Dataset Diversity Analysis

To further evaluate dataset diversity, we analyze multiple complementary aspects including low-level signal attributes, duration distribution, resolution distribution, and semantic category coverage.

(1) Low-level audio-visual attributes.

To further evaluate dataset diversity, we analyzed several low-level audio-visual attributes and compared their distributions with representative AVQA datasets. Figure 10 illustrates four representative attributes, including video spatial information (SI), temporal information (TI), and audio spectral centroid (SC) and spectral entropy (SE).



(a) Temporal Information (TI)

(b) Spatial Information (SI)

(c) Spectral Centroid (SC)

(d) Spectral Entropy (SE)

Fig. 10: Distributions of normalized low-level audio-visual attributes across four datasets: UnB-AVQ, LIVE-SJTU, SJTU-UAV, and our YT-NTU-AVQ. Our dataset demonstrates broader coverage in both video (TI, SI) and audio (SC, SE) domains.

Although these attributes were not explicitly controlled during sampling, our dataset maintains distributions comparable to the UGC-based dataset SJTU-UAV, indicating good coverage of low-level audio-visual characteristics. Compared with earlier AVQA datasets, our dataset also exhibits broader coverage across both video (TI, SI) and audio (SC, SE) dimensions.

(2) Duration consistency.

To facilitate processing and evaluation, A/V sequences are extracted with a target duration of 10 seconds, consistent with AudioSet. Due to source video boundaries and trimming constraints, a small portion of clips deviate slightly from the target duration. In practice, the average duration is 9.91 seconds with a standard deviation of 0.59 seconds, and 92.5% of clips fall within 9.8–10.2 seconds. This high consistency ensures stable subjective evaluation and model training.

(3) Resolution distribution.

The dataset contains a wide range of spatial resolutions, including many non-standard formats commonly observed in real-world UGC content, reflecting realistic diversity in capture devices and recording settings. The spatial resolution distribution is illustrated in Figure 11.

(4) Semantic category distribution.

To analyze content scenario diversity, we define 9 coarse semantic video content categories based on dominant visual and contextual content. These categories are designed to reflect common audio-visual interaction scenarios rather than fine-grained semantic taxonomy. Video-level labels are assigned using Gemini 2.5 based on primary content description.

The semantic category distribution is shown in Figure 12.

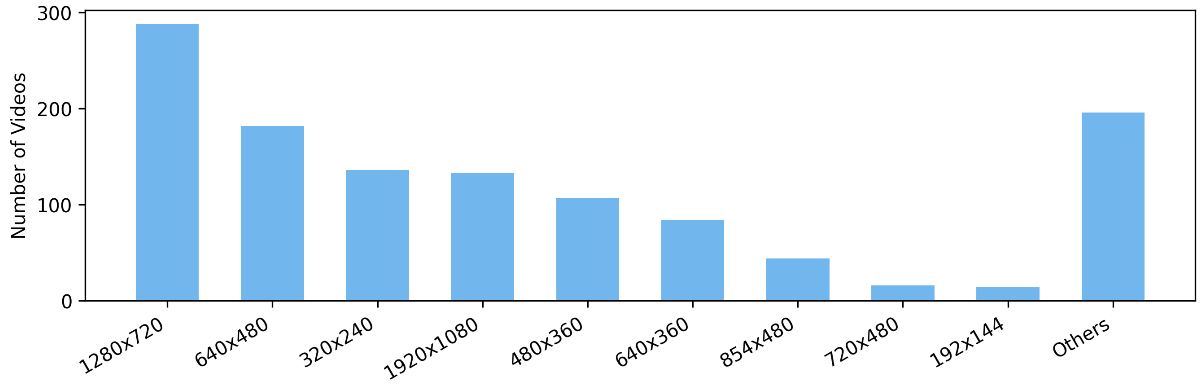


Fig. 11: Distribution of spatial resolutions in YT-NTU-AVQ. The most frequent resolutions are shown individually, while the remaining resolutions are aggregated into an “Others” category.

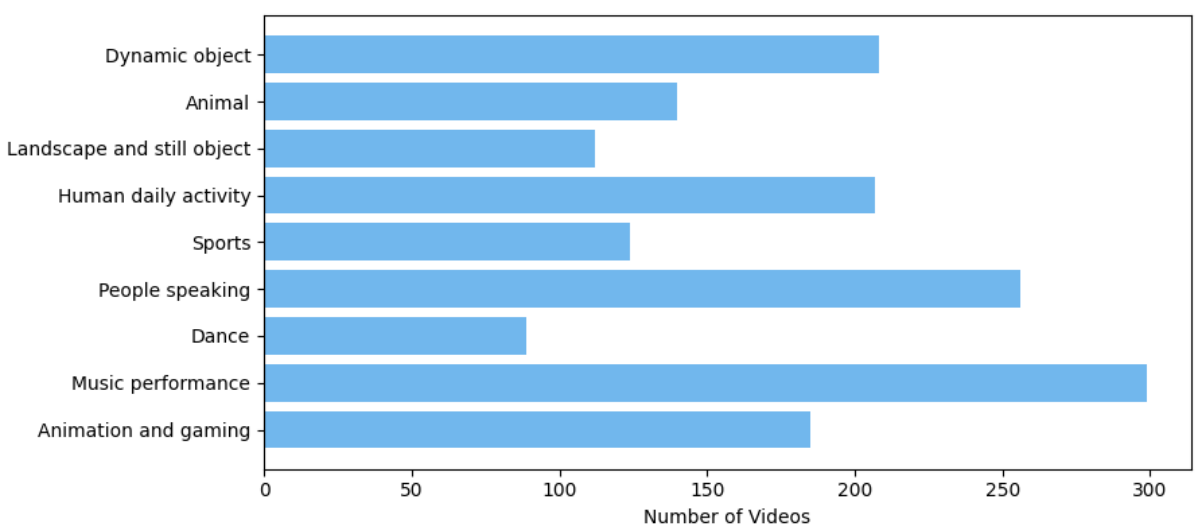


Fig. 12: Distribution of videos across predefined coarse semantic content categories in YT-NTU-AVQ.